

Selecting the Right Blockchain Technology

As blockchain continues to evolve, so do your opportunities. This article is a roadmap to informed blockchain decisions and will help you navigate public, private, and hybrid options with precision.

A blockchain is a decentralised and distributed digital ledger that records transactions across multiple computers in a way that ensures security, transparency, and immutability. Understanding the distinctions among various blockchain platforms is crucial when embarking on a blockchain project. Before diving deep into the crux of selecting the blockchain platform for your project, let us recap some fundamentals of blockchain and its types.

Brief overview of how blockchain works

While venturing deeper into the blockchain landscape, it's essential to understand how this technology's inner workings shape the choices in the world of decentralisation. Here is a brief overview.

Blocks: Transactions are grouped together into blocks. Each block contains a list of transactions and a

reference to the previous block, forming a chain of blocks.

Decentralisation: Instead of being stored on a single central server, the blockchain ledger is maintained by a network of computers (nodes) distributed around the world. This decentralised nature prevents a single point of failure and makes the system more resilient.

Consensus mechanisms: To validate and add new blocks to the chain, a consensus mechanism is used. The most well-known consensus mechanism is proof of work (PoW), which requires miners to solve complex mathematical puzzles to verify transactions. Other mechanisms like proof of stake (PoS) and delegated proof of stake (DPoS) exist as well.

Immutability: Once a block is added to the blockchain, it's nearly impossible to alter the information within it. This immutability is achieved through cryptographic hashing and the decentralised nature of the network.

Transparency: The entire transaction history is publicly available on the blockchain, offering a high level of transparency.

Security: The decentralised and distributed nature of blockchain, along with cryptography, enhances security. Changing a single block would require altering subsequent blocks on every node, which is computationally infeasible.

Blockchain use cases

Blockchain technology has transcended its roots as the backbone of cryptocurrencies, and has evolved into a versatile tool with myriad applications. It now has applications beyond cryptocurrencies.

Supply chain management: Blockchain can provide transparency and traceability in supply chains, verifying the origin and movement of goods.

Smart contracts: These are self-executing contracts with the